

Securities Education Certificate



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Lecture 4

Derivatives I

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Bonds (A Primer)

Before we discuss derivatives in great detail, I did promise that this topic would be covered in Lecture 6. However, since bonds are such a key critical part of the world of finance, it seems to find itself over and over again in the SEC whenever we mention the risk-free rate and things like that. So, what I'm going to do here briefly is explain what bonds are and a brief explanation as to how they work. Don-Angelo in Lecture 6 will cover them to a much greater detail than what I have here – I've only done the brief details so that you can understand the next two lectures without being confused and/or getting annoyed that we *promised* it'll come in Lecture 6.

Equity and/or Debt – what's the alternative?

I think by this point, the picture is quite clear. If you own a company and you want to raise funds to kick-start or continue the company's operations, you're either going to sell shares in the company or take out a loan in the form of a debt.

There are, however, two immediate issues when taking either of these approaches:

- Selling equity means selling ownership of assets. In turn, since income can be converted into cash (which is a business asset that shareholders can claim), this results in shareholders staking a claim into the company's profits through e.g. retained earnings. If you are the owner of the company, you could lose out on potential future profits if you choose to sell shares in order to raise funds.
- Securing a loan i.e. debt results in you having to pay interest payments on the debt. This would of course mean that the net income of the company decreases and hence the company makes a smaller profit.
 - ◆ Remember, the interest rate is **NOT** determined by you but rather by the *lender* of the debt. They choose the percentage rate that you have to pay back on the debt every year, so you have no room to negotiate on those matters.

You can sort of see the problem: either you give away ownership of future company profits or you get shafted by the interest rate you are charged on a loan.

However, there is a third way!

What if there was a way where you could raise funds from third parties **WITHOUT** selling any equity in the business and you could set the interest rate yourself on the money you are given? Wouldn't that be great news for you since you're the one in control of the lending of cash to yourself!

That is a bond.

A bond is essentially just a loan that an investor makes to a borrower (e.g. a company or an individual) *at the terms of the borrower*, **NOT** the lender. The borrower sets the terms and the investor has to meet those terms if they want to loan the money. It's as simple as that! Often, I like to think about the word 'bond' as a *binding* agreement between the lender and the issuer although in this instance, the issuer is the one that is in the driving seat when it comes to lending negotiations.

Let's think about this a bit in terms of the actual underlying mechanics a bit more so that we can really see what's going on under the bonnet.

How does a bond work?

Let's start with this country's biggest and most famous company... the UK government. Some of you are going to be thinking "how on Earth is the UK government going to be considered a company?" Really, it's not (probably) but it's good to think of it as one. Think about it: they earn money through taxes and other business that it does and spends that money on projects, teachers and lots of other interesting stuff that gets them on the front cover of the Daily Mail every single day.

Sometimes, the Government doesn't have the money to e.g. build a new bridge. What the Treasury (i.e. the business team of the Government) does is that it goes around to different potential investors and says "We're looking to issue bonds in order to finance this bridge. We need £1 billion, so we're going to issue 1 million bonds where each of you investors pay us £1000 and we'll pay you back 3% every year for 10 years. When the bridge is built after 10 years, we'll pay you back that £1000 after those 10 years."

That is a very simple and easy way to understand the bond. That's all it is and that's all that it'll ever be. Bond issuing often happens with the Government when they are experiencing a **budget deficit** i.e. they haven't got enough money coming in to pay out all the projects taking money out of the government. Yes, they could go and get one massive loan from Wonga, but why would they? There are keen investors who think the Government is a safe bond issuer.

One misconception I'd like to clear up is that Governments are not the only entities that can issue bonds. Large firms e.g. big industrialised companies can do so. There's a lot of legal broo-ha-ha about what companies can issue bonds in terms of credit ratings (i.e. how likely they are to pay back the money) but that's a story for another Lecture. The bottom line is that governments *and* firms can issue bonds as much as they please.

Right. Onto the mechanics of bonds!

Setting up a bond

Let's look back at that example earlier.

- Initial payment £1000
- 'Interest Rate' 3%
- Duration of bond 10 years

Simple stuff. In 'Bond Theory', these terms have unique names:

- **Bond Principal** £1000
- **Coupon Rate** 3%
- **Maturity Date** 10 years (from issue)

Rather than leaving you with the definitions and running away, I'll cover brief definitions of each term so that you're safe in the knowledge that you know what you are learning about!

- The *bond principal* is the amount of money the lender pays to the borrower. An alternative definition you'll see is the amount of money *an issuer is borrowing* by the terms of the bond. Sometimes it's known as the 'nominal value' i.e. the number that you see printed on the front of the bond when it comes out. It's the same thing, honestly.

There are two parties at play here: the borrower is the *bond issuer* and the lender is the *bondholder*. If

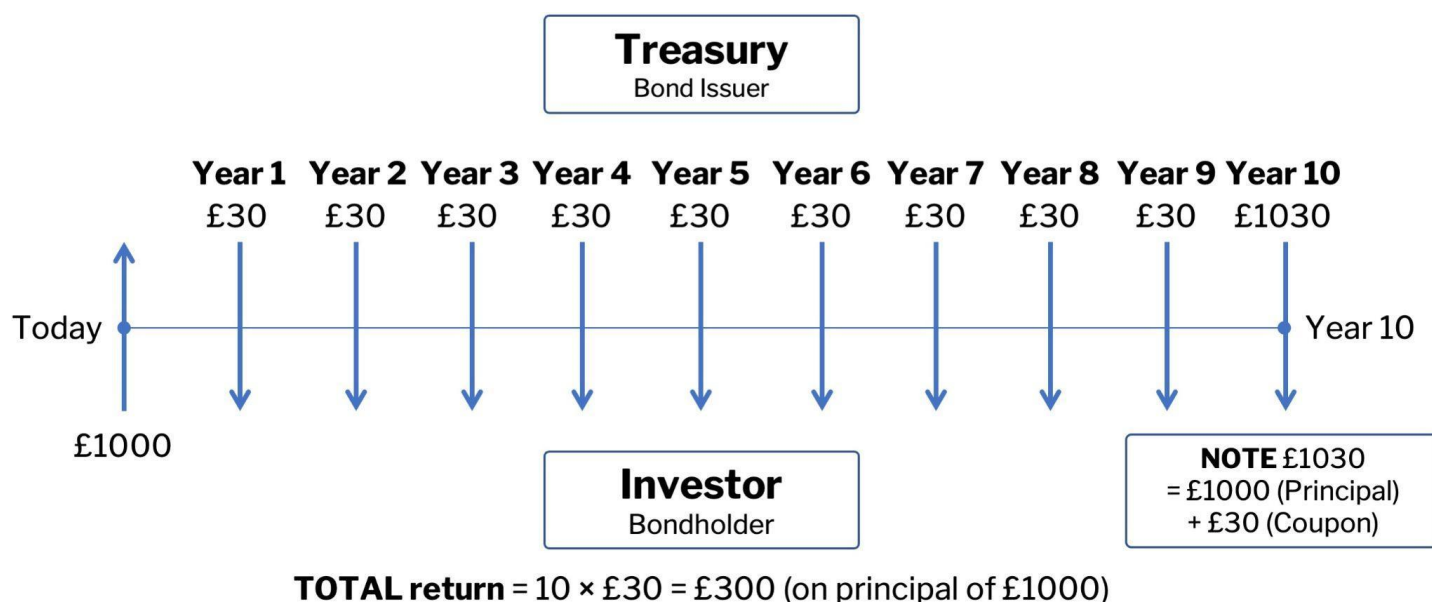
that doesn't make sense, essentially the bond issuer creates the bond contracts and the bondholder is the one who pays into the bond and holds the contract until it expires.

- The coupon rate is simply the interest rate paid on the bond. In this case, it's 3% of the *bond principal* i.e. £30. Often, this coupon rate is paid annually i.e. every year. Sometimes, it's paid *semi-annually* i.e. every half a year (6 months).
- When a bond is paid off and all finished, we often say that the bond has **matured** i.e. grown up. Once a bond is matured, the principal's been paid back to the lender with the coupon payments and everyone is happy. Sometimes, we say the bond *expires at maturity* – literally, it means the exact same thing.

In the olden days, a bond would be issued to the bondholders through a fancy certificate with coupons attached on the bottom of the page. Once a year would pass, the bondholders would snip off a coupon and travel to the bond issuer to demand coupon payment. That's where the idea of a *coupon* essentially comes from!

Picturing the bond's cash flows

Really, this is best done with a diagram. Let's try one.



The top half of this diagram shows the bond issuer and the bottom half shows the bondholder. The bond holder delivers the principal of £1000 to Her His Majesty's Government. In return, the Treasury pays £30 to the bondholder every year for 10 years, earning a nice sum of $£30 \times 10 = £300$. After 10 years have passed, the bondholder receives the final coupon of £30 along with the original £1000 he paid in the first place. 'Cash money.'

Sometimes, bondholders like to know how much money they're making every year as a percentage of the principal they donated i.e. their *return*. This is known as the **yield** of the bond. Think of a farmer: every year, they plant seeds and they get a certain percentage of the seeds converted to good crops. Essentially, the same exists when it comes to bonds: the bondholder planted an initial principal on a government or firm and in return, they get a certain percentage back every year.

The formula is given on the top of the next page for those interested.

$$\text{Bond Yield (\%)} = \frac{\text{Coupon Rate}}{\text{Market Price}} \times 100$$

So, for a coupon of £30 on a bond of £1000, the yield is 3%.

Some of you might ask “well, why would I need to know anything about yields? They are the exact same values as coupon rates.” And you’d be right... until you realise that bonds can be bought and sold for any amount of money i.e. they’re traded on an exchange. As a result, the yield that an investor makes is actually the coupon rate expressed as a percentage of the market price they paid for the bond.

For this reason, we specify the market price and not the principal of the bond when calculating yield. It’s certainly valid that as soon as the bond has been issued, the principal and the market value are identical as people are willing to sell £1000 of their own money to Rishi Sunak’s new government but after a while, those bonds might be traded at higher or lower prices. For this reason, bond yields can change given that the denominator is the only thing changing on the right hand side. How this actually happens is the story of Lecture 6 so I won’t take too much away from Don-Angelo Sfeir as he’ll be the one delivering an excellent lecture, that’s for sure!

I do hope you’ve taken a brief insight into the world of bonds here using a bond from the Government with coupons attached. One thing to bear in mind is that Government Bonds (with the UK bonds uniquely named ‘gilts’) are considered to be the *safest* investment in finance since it’s unlikely a Government can just run out of money to pay off its bondholders – a simple tax hike can fix anything. Don’t however be fooled by the rate of return on a bond being coined a ‘risk-free return’ – there is certainly a finite chance that Governments can go bankrupt although very few have... yet.

On those last and rather foreshadowing words ***knocks wood***, let’s crack onto derivatives.

“Financial weapons of mass destruction that, while now latent, are potentially lethal.”
 – Warren Buffet (2002)

Derivatives

Introduction

Before anyone is fooled, myself and Oscar can assure you that derivatives are inherently an abstract idea which might seem completely off-putting to learn at first. However, once you realise why we need them, you can then have a better understanding as to how they work and the motivations behind why an investor would choose to work with derivatives through their career. That’s how we plan to teach this lecture: we begin with the ‘why’ and then work towards the ‘what’ with regards to derivatives. Here, we only discuss the preliminary basics since there is so much more to derivatives, with a lot of maths surrounding their behaviour if you choose to study the more advanced types of derivatives. So, without any further ado, let’s crack on!

The first thing I want you to understand is this: if there is money to be made, then anything can be arranged. When you read these notes and you think “why on earth would anyone want to arrange such a deal?!” , remember that phrase... either a buyer or seller will win in any deal!

So far, you’ve got an idea as to what financial markets are all about (Lecture 1), who is involved in them (Lecture 1 + 2), what drives the markets on a daily or annual basis (Lecture 2) and what sorts of assets people can purchase (Lecture 3).

Now, it’s not a huge surprise that people that work in the financial markets have one objective only... *make as much money as you can and lose as little of the firm’s capital as possible*. There are many ways that financial players (or actors, however you wish to call them) can invest their cash to maximise their returns, but any investor will tell you there is a possibility that their investments could fall in value and hence they’ll recoup less money back if they choose to sell the assets they acquired – who knows, a freak environmental disaster or an economic recession could happen in the near future.

Definitions

Before we discuss the idea of derivatives, it’s important to define a few rigorous terms associated with the bad outcomes that could arise from an investment.

- A **risk** (n.) is simply the possible *negative* outcomes that arise from a given event happening. A risk is often measured by two things:
 - How likely is the event to happen?
 - How bad are the effects going to be as a result of the event taking place?

A simple example – a homeowner defaulting on their mortgage. Banks do a background check on a customer’s credit history to see if they have a history of paying back *credit* loans on time and hence minimise the risk of the loan not being repaid (this is frequently referred to as a **default** [n.]). If they also don’t pay back the loan on the house, the bank repossesses the house and sells it to try to recover the money owed. If the housing market is plummeting as well, they probably won’t be able to recover the full value of the loan that they sold to the customer for the house to be purchased, which in itself is another risk to the bank. Notice here that the effect on the bank is that they’ll be out of money and have to ‘balance the books’ to stay afloat, which was a huge issue back in 2008 when the housing market fell apart as the banks issued thousands of NINJA loans (No Income, No Job, no Assets) and were

surprised that the customers in question couldn't pay back the loans they'd taken out... What a surprise!

- A business deal can be simplified to a buyer and a seller (as we all probably are aware...). It's better to think of both of them as two pieces that together form the complete puzzle of a sale. A **counterparty** in a deal is either of these two pieces of the deal i.e. either the buyer or the seller. This might not make sense from the formal definition, so the following example will help alleviate the definition's abstract nature. If you are the buyer, the party on the other side of the deal is your counterparty i.e. the seller. Likewise, if you are the seller, the buyer is your counterparty in the deal. The counterpart of a deal that you're in is – quite literally – the counterparty. Why might this be relevant? Often, you will hear the concept of **counterparty risk**. Google defines it as *the likelihood or probability that one of those involved in a transaction might default on its contractual obligation*. Simply put, it's the risk that the party on the other side of the deal refuses to honour the deal and deliver the goods or the money, based on whether they are the buyer or the seller. Counterparty risk forms the basis of the motives behind derivatives.
- To **hedge** (v.) a risk is to take precautionary action so that if a bad event occurs, there is a way for the risks to be minimised. Airlines often buy aviation fuel a year ahead of time at current prices so that if the fuel prices increase a year from now, they can load fuel into the planes in 12 months time at the cheaper prices offered today, hence minimising the risk of having to pay increased prices (how convenient for today's world, eh?). Conveniently, the same idea is incorporated into the term '**hedge funds**' – you'll come onto them at a later date if you choose to attend one of the society's career talks in the coming weeks.
- When you invest in an asset, you *gain* something called **exposure** (n.). In short, exposure is basically how much capital you stand to lose if your investment fails. If you pay £1 million for a house and tomorrow the housing market collapses, your exposure is exactly £1 million because tomorrow your house could be worth £0. Some exposures can be *greater* than the price you pay for a given asset (have a guess how this can be possible... **negative pricing**) – this is often the case in commodity pricing, which you will see in Lecture 7.
- The **spot** price of an asset is how much someone is willing to buy the asset for **right now** i.e. on the *spot*. The spot price of an asset can be found on exchange boards. For example, you could go and buy Tesla stock right now for a spot price of \$622. This price represents the amount that a trader will pay for instant delivery of the asset (in this case, a stock) and will constantly fluctuate depending on supply and demand. The spot price is largely responsible for determining the price of a derivative as we will discover later. Notice here the price of the stock is quoted in U.S. Dollars (USD) – that's because Tesla stock are traded only in America at an exchange called the NASDAQ (the National Association of Securities Dealers Automated Quotations – these acronyms really are bizarre at times).

The what and why of derivatives

Lovely. So, we now know that all investments have an intrinsic risk (by intrinsic, I mean a built-in, natural risk that we can't avoid – we buy an asset for a price that by definition can lose us money if its value falls to 0). We want to hedge these risks so that we don't lose any money, which will help us minimise our exposure to capital loss if things go awry. Thankfully, for hundreds (if not thousands of years), investors have developed specific contracts containing mechanisms that help them hedge risk and minimise exposure whenever they invest... This is what a derivative does. In mathematics, derivatives help you calculate the rate of change of a function $f(x)$ when evaluated at a point x . In finance, derivatives help you manage risk and minimise exposure alongside a few other niche benefits – they are NOT the same by any stretch of the imagination, although you might notice they are named the same because of the *underlying* idea.

Derivatives answer the following question for investors:

How can someone hedge their risks and therefore minimise their exposure to an investment?

'Derivatives' is an umbrella term for a range of different types of contracts that offer the possibility for an investor to hedge a risk or make money from something called '**speculating**' – believing that an asset is going to go up in price in the future. We should be aware of what a contract is but for those that aren't sure, a contract simply involves two or more different 'parties' (which can be people, institutions etc.) and binds them to an agreement with specific terms and conditions attached. **A derivative is a contract of which its value is based on an underlying asset.** More generally, it gets its value from what is otherwise known as a *financial product*.

The underlying asset (often denoted as UL – **UnderLying**) can be confusing at first to wrap your head around, but it's quite simple to understand with some real-life examples. Most of you have come across derivative contracts based on an underlying asset before. When you buy insurance for your car or home, the insurance company quotes you a specific price based on the condition of the car or the location of your home. The car and the home is the underlying asset here – it's an asset that's taken into consideration by the insurance company to sell you a contract promising protection and cover if it crashes or goes up in flames (touch wood). Alternatively, if you don't own a car or home, you might be familiar with a sports bet. If Arsenal are playing Manchester United on Sunday, a bookmaker or betting shop might quote you betting odds based on the result of the football game once it's played. They might offer you 10/1 odds if you think Manchester United will win 3-0 but instead offer 2/1 odds if you think Arsenal will win 1-0 (although we don't advise you to get bogged down on how betting odds work – just the idea that the predicted result of the game is used as a basis to quote odds and prices is the point of this example). It's a good idea to first think of options as bets although you will find that idea will become more blurred when we discuss specific derivatives later on, such as futures and forwards (but hey, it's a good start either way.)

There are a wide variety of underlying assets that can be used to form the basis of a derivatives contract, including stock indices, bonds, interest rates – even the weather conditions could be the basis of a derivatives contract. You might now start to realise how different underlying assets might be used by an investor to hedge risk at this point, but if you don't, you'll find out later once we get into the specifics.

Note: Some of you might have realised, for instance, that we compound the concepts of the weather and interest rates with the idea of an underlying asset. You'd also be right to question this idea... Of course, you can't own the weather nor can you own the interest rate. However, if you are e.g. a farmer, you would grow crops of which its value depends on the state of the climate in the growing season. If you base a derivative off the price of a crop (which is linked to the weather, which in turn influences supply – Lecture 2), you would have a derivatives contract that is indirectly linked to the weather in the end anyway. Likewise, treasury bonds (see Lecture 6) also are linked inversely with interest rate, so if the interest rate set by a CB falls, treasury bonds increase in price and then you can have a derivative linked to the price of a treasury bond, which depends on the CB interest rate! As a result, you have a derivative indirectly linked to something you can't really own but want to monitor to hedge your risk. I personally dislike the idea of a UL and instead like to think of a 'contract driver' to eliminate all ambiguity as to what really affects the derivative itself, but it's all personal preference in the end.

What types of derivatives exist in financial markets?

There are several types that you will come across if you choose to work in financial markets. We'll only be covering a few of them, which you see listed below in the order that they're featured in the notes:

- Futures
- Forwards
- Options
- Swaps

Shorting

You may have heard the terms selling and shorting used interchangeably, but there is in fact a subtle difference. Selling an asset involves exchanging an asset **that you own** for an agreed payment. Shorting involves the process of **borrowing an asset and selling it** on the market for cash now with the hope that you can buy the same asset back later at a cheaper price and return it to the lender. An example of this kind of trade is,

1. Borrowing a share of a company worth \$100 now that you think is going to go down in price.
2. Selling it immediately on an exchange to gain \$100.
3. At a later date, buying the same share back for a lower price of \$90.
4. Returning the asset back to the lender and pocketing the difference between what I sold it for and what I bought it for (\$10 profit)

Of course, the price of the share could go up in value and we would instead make a loss.

Because none of your own capital is used to borrow, the only money required to initiate a short trade is a small amount that acts as margin to the lender of the asset. margin is an amount of money held to cover the risk of you not being able to buy the asset back. If you make a loss it is taken out of the margin account and if the stock price stays the same, you get your margin back when the trade is closed. As only margin is used, traders can leverage their short positions by e.g. putting down \$10 to have exposure to price movements in a stock worth \$100. If the stock price increased by 10%, their total profit would be $\$110 - \$100 = \$10$. This is a 100% increase from the \$10 initially provided. By putting down only a small amount of money, but still getting all of the profit and loss on a much larger underlying value, they are creating **leverage**.

Shorting is a part of most FX trades done to hedge or speculate - more detailed diagrams of this can be seen in the slides.

Feel free to write your own notes here. The content on futures and forwards begins on the next page.

My notes

Types of derivatives

Futures and Forwards

To get a basic understanding of derivatives we can start with the two simplest types: **futures** and **forwards**. The difference between these two derivative contracts is essentially that *futures can be bought by anyone, forwards can only be bought by institutional investors dealing with a trader at a bank*. This will be discussed later, but they function largely identically so for now we will look at both.

Both futures and forwards contracts allow someone to **buy or sell** (depending on if they are long or short the contract, respectively) **a set amount of an underlying asset, for a set price at a set point in the future** (e.g. you have to buy 100 barrels of oil for \$100 per barrel in 3 months time, referred to as the *expiry* or *settlement date*). At the agreed set point in the future, the buyer/seller is obliged (i.e. they have to do it) to buy or sell the agreed amount of an underlying asset for the agreed price. Note that goods and cash are only exchanged at the settlement date. One of the parties to a forward or futures contract assumes a *long position* and agrees to buy the underlying asset on a certain specified future date for a certain specified price. The other party assumes a *short position* and agrees to sell the asset on the same date for the same price. It can help to look at an example to get a better understanding.

Let's consider an example of a futures contract in the context of agricultural livestock.

> Example 1 - Futures and Forwards

You enter a futures contract for live cattle that expires (or 'reaches the agreed transaction date') in 2 months for \$53,500 (this price is roughly accurate and will get you 50,000 pounds or around 60 live cattle!). 2 months pass and the spot market price (i.e. current price) of cattle is now \$60,000. If the cows are sold immediately after delivery, *what is your profit/loss?* To be clear here, when we mention delivery in finance, we are talking about the cows being transferred i.e. delivered from the seller (probably a farmer in this case) to you, making it your responsibility to look after them however you wish once the contract's been agreed.

To answer these sorts of questions, let's look at the question from the perspective of our cash flow.

CASH FLOW STATEMENT (2 MONTHS)	
Event	Cash Flow (\$)
Today	No money is transferred and the contract is simply entered, not executed yet.
2 months later	– 53,500
We buy the cattle 2 months after entering the contract.	<i>Because we are bound by the contract to do so.</i> In cash flow statements, this is denoted as (53,500) i.e. with brackets around the number to denote (-)
The cattle are then sold at the spot market price of \$60,000.	+ 60,000

TOTAL PROFIT	+ 6,500
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Table 1.1 - Example 1

If you think this example was pretty easy, you'd be right – because it was. However, more complicated situations require intricate analysis of each thing involved so breaking it down in an appropriate structure saves you lots of stress, time and your proprietary trading account. Of course, this could go the other way and the spot market price of cattle could go down leading to a loss. If supply of cattle increases whilst demand does not increase (or even decrease), the price could then fall to e.g. \$40,000 and you'd be facing a loss of – \$13,500 (that is, $-53,500 + 40,000 = -13,500$). Like we mentioned before, derivatives are great when used right to give you huge profit. but dangerous when you're not careful of their risks.

So, the above example makes sense but why would anyone use a forward or futures contract instead of buying the underlying asset straight away? They have the same outcome, right? Either way, I get to own 60 live cattle after paying the appropriate price(s) involved. **In theory, yes you're right.** But, do you really want to look after oil/cattle/gold for 2 whole months? Think about it – if you buy cattle today, where are you going to store it? You might impulsively buy the cattle today as you believe that today's price will be the cheapest for the next 12 months, but where do you store it? Do you own a farm to place all these cattle? If you're a trader, you probably don't. Can you feed them for 2 months before selling them on at a higher price? Probably not, especially if you need to fund their food and other associated costs (you don't want to sell cattle that's underfed or even worse, dead). Futures and forwards avoid this issue by allowing you to lock-in i.e. fix a certain price for something you need in y months time in the future without the need to purchase it now, paying high rates of storage costs to store it for several months before using the asset itself.

Where are forwards and futures used? The aviation industry makes heavy use of futures and forwards to fill up their planes with aviation fuel. Imagine that you own an airline and don't want to risk the price of oil going up over summer when you have a large amount of flights to run. You could either buy jet fuel now for $\$x$ and have to store it until summer, or you could buy a forward contract. This allows you to fix the current price of jet fuel and not have it delivered until summer when it's needed. Jet fuel melts steel beams (probably), so you wouldn't want to put it in a warehouse for a year or two before you plan to use it because if it catches fire, the only place your investment is headed is into the sky in a puff of smoke and you certainly can't pump that into a Boeing 737 to make it fly.

As a trader, you may also prefer futures/forward to buying the underlying asset as you don't actually have to keep the asset but instead to benefit from its price movements in addition to being able to access large amounts of leverage unavailable when buying solely the underlying asset. You'll see this being mentioned further in these notes, but there are possibilities in finance where you can enter a derivatives contract without actually having the underlying asset delivered - this is called **cash settlement**. Essentially, you would own the UL for a set period of time and profit from any change in price when the contract expires. There's more on this later, so if you don't understand now then do not worry!

Let's take an example for starters. Oil is a common topic when it comes to derivatives, so let's add some realism to our notes. For example, the image below shows the January expiry oil futures contracts on one of the most popular derivatives exchanges, the Chicago Mercantile Exchange (CME).



Figure 1 - Crude Oil Futures Contract (Jan expiry)

If you go long on this contract the price at the current time is locked in. At first no money is exchanged, and then on the 2nd of January, you pay the seller the quoted price upon entering the contract and in return get 1000 actual barrels of oil, or more commonly a cash settlement equivalent to the profit you would've made.

Remember, longing a contract means you're in a position to buy it but shorting a contract means you're in a position to be selling it. This is of course not ideal if you don't actually want a physical barrel of oil so most contracts are **cash settled** i.e. you instead get the difference between the price you are forced to pay (strike price) and the current price of oil.

Notice how the quoted contract price fluctuates over time, this can be attributed to changes in the underlying price of spot oil (i.e. oil for delivery now) as well as how long the contract has until it expires. We will cover this more closely later but it's good to realise that this type of derivative can be actively traded just like a stock or commodity.

How do futures and forwards differ?

As promised, we will now look at the difference between the seemingly identical futures and forwards. The main difference is how the two contracts are traded. **Forwards are traded over the counter (OTC) and futures are traded on an exchange.**

Let's start with futures, and how an exchange actually is involved.

Exchange traded markets - Futures

Starting with the type of trading that you may be familiar with, exchange trading. A derivatives exchange is a market where individuals can trade contracts that have terms defined by the exchange - standardised contracts. This means that everyone on the exchange will have access to a basket of the same type of contracts (e.g. 1000 barrels of oil expiring in September may be one standardised contract or expiry in October may be another) and so there are generally lots of buyers and sellers for the same contract. This creates what is called liquidity; how easy it is to unload (buy or sell) a given contract. You can see this in Figure 1.1 above, there are specific contracts that can be traded rather than tailored products.

Initially, the Chicago Board of Trade (CBOT) was created in 1848 to bring farmers and merchants together with a main focus of standardising the quantities and qualities of grains being traded. Farmers used to trade any quantity of a commodity e.g. 1, 32 or 69 tonnes of wheat and potatoes. This became an issue as buyers wanted a more consistent system where they could be quoted specific quantities of a commodity e.g. multiples of 100 tonnes, which is what **standardisation** helped to achieve. Not long after, the CBOT introduced the first futures-type contract, the *to-arrive contract*. This is indicative of the scheduled delivery date of a set amount of standardised grain. The CBOT no longer exists and instead merged with the Chicago Mercantile Exchange to form the CME Group which is one of the largest derivatives exchanges in the world.

Let's go back to the mechanics of the exchange. Once two traders have agreed on a trade, it is handled by the exchange clearing house. The **clearing house acts as a middleman between traders** and manages the risk to each party. This ensures that when trading on an exchange, **each trader does not need to worry about the creditworthiness of the other trader**. This is possible as the clearing house requires each party to provide collateral in the form of 'margin'. **Margin** is a set amount of money that each trader has in their trading account that can be used to pay any debts which may arise when a trade goes against them.

Let's try another example to make this clear. If Ahmad enters a futures contract on an exchange for oil, the exchange quotes he may receive are below:

MONTH	OPTIONS	CHART	LAST	CHANGE	PRIOR SETTLE	OPEN	HIGH	LOW	VOLUME	UPDATED
 SEP 2022 CLU2	 OPT		98.30	+1.88 (+1.95%)	96.42	97.30	101.88	96.41	331,035	16:37:58 CT 29 Jul 2022
 OCT 2022 CLV2	 OPT		96.41	+1.76 (+1.86%)	94.65	95.42	99.75	94.59	109,019	16:37:55 CT 29 Jul 2022
 NOV 2022 CLX2	 OPT		94.96	+1.68 (+1.80%)	93.28	94.08	97.91	93.23	70,414	16:38:30 CT 29 Jul 2022
 DEC 2022 CLZ2	 OPT		93.40	+1.49 (+1.62%)	91.91	92.68	96.15	91.82	94,427	16:38:25 CT 29 Jul 2022
 JAN 2023 CLF3	 OPT		92.00	+1.41 (+1.56%)	90.59	91.38	94.50	90.59	25,981	16:37:45 CT 29 Jul 2022
 FEB 2023 CLG3	 OPT		90.98	+1.58 (+1.77%)	89.40	90.22	93.09	89.47	13,432	16:38:07 CT 29 Jul 2022

Figure 2 - Futures exchange snapshot

Above, you can see a set of different futures contracts for West Texas Intermediate oil (you can also get Brent Crude and Dubai oil as you saw in the commodities lectures) traded on the Chicago Mercantile Exchange. Along the left edge, you can see the different expiry times for each contract. Looking across to the right you can then see the Last price obtained (i.e. last trade to happen), the Prior Settle (i.e. the price traded at the end of the last trading period), and the open, high and low. The open price is the price of the contract at the start of the current trading window, high is the highest value trade that occurred in the window and low is the lowest value trade that occurred. The volume gives the number of each contract that was traded in the day. Each of these contracts has a price which represents the price per barrel, however each contract is specified for 1000 barrels. So if you entered one of these contracts, your exposure would be the price x 1000.

Let's say he picks the Dec contract, so the settlement occurs on the 14th of December. Each contract is for 1,000 barrels and he 'locks' in a price of roughly \$93.40 by entering this contract (assuming that he can get the contract for close to the last traded price). His exposure is therefore \$93400.

Exchanges typically ask for a 3-12% margin depending on the volatility of the underlying asset and so Ahmad would have to give \$2,802 - \$11,208 as margin to ensure that, if the trade goes against him, there is money that can be used to settle the trade debt. Remember, volatility is how much the price of an asset tends to change in a given time period so we often use the standard deviation to calculate it.

The image above of futures price quotes shows an electronic exchange market, where trades can be agreed and entered through a computer. Traditionally, derivatives exchanges have used an *open outcry* system. This required traders to physically meet on the trading floor or just shout or signal trades that they would like to complete across the room. Exchanges have now largely been moved from this system to electronic trading for one main reason: it's much faster and less likely to go wrong, making sure all the participants are 'heard' equally online.

Over-the-Counter Markets (OTC) – Forwards

These are the markets which you will not see until you're working in finance yourself. Only the biggest players such as banks, other large financial institutions, fund managers and corporations are the main participants in the OTC derivatives markets. This market primarily consists of bank traders quoting prices for certain financial instruments that the bank may offer, often described as 'bespoke' financial products.

Traditionally, participants in the OTC derivatives markets will contact each other directly by phone and email, or will have counterparties for their trades using an interdealer broker. Here the participants (perhaps a hedge fund looking to speculate on the price of gold) will be able to phone up their designated trader at a bank and get a quote on the bid (short position price or 'how much the bank will pay') and offer (long position price or 'how much the bank will sell for') for a particular forward contract. Banks often act as market makers for the more commonly traded instruments.

Once an OTC trade has been agreed, the two parties can either settle (execute) the trade with a central counterparty (a CCP – basically, just an intermediary between the two parties) or clear (ensure the trade happens) the trade bilaterally. A CCP stands between the two parties to the derivatives transaction so that one party does not have to bear the risk that the other party will default. Alternatively, when trades are cleared bilaterally, the two parties have usually signed an agreement covering all their transactions with each other. This agreement covers most importantly, how much collateral must be posted by each side of the trade and under which circumstances the outstanding trades can be terminated.

How big are these markets?

Huge! Well, at least the value of the underlying assets which derivatives are based on are. For example, an over-the-counter agreement may be to buy 100 million US dollars with euros but typically only 1 – 2% of this amount would ever change hands as the cash flows cancel out and only the profit is transferred from the trader who lost on the trade. The frequency of OTC derivatives trades is smaller than the number of exchange derivatives trades. This does make sense when you consider the ease of trading electronically and high frequency trading compared to phoning or emailing a bank to ask for a quote on an OTC-traded forward. However, the average size of an OTC trade dwarfs the average exchange trade alluding to a much larger OTC derivatives market (by market cap or total underlying assets) than the exchange traded derivatives market. This also makes sense if you consider why you wouldn't want a computer to be handling deals worth tens of millions or billions of GBP each.

The figure below compares the estimated total principal amounts underlying outstanding transactions in the over-the-counter and exchange markets between June 1998 and December 2021. This demonstrates the explosion of derivatives into the 21st century. Around 2008, you can see a decline after the financial crisis

caused primarily by the danger of leveraged derivatives. Today, the total derivatives OTC market is estimated to have an underlying principal (assets combined value) of over \$1 quadrillion!

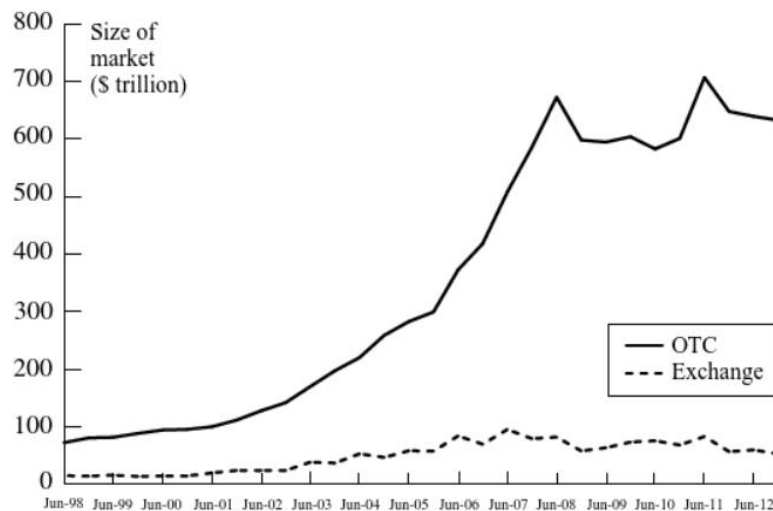


Figure 1.1 Estimated size of underlying assets for derivatives markets

The main reasons for the huge success of the derivatives market include the ability for traders to access **large amounts of liquidity** (lots of buyers and sellers) as well as the **wide range of different payoffs** (profit and loss behaviour) that can be synthesised with just one underlying asset.

How are these contracts normally traded?

Imagine if every time that you entered one of these derivatives contracts, you had to take delivery of the underlying asset. As one junior trader in a large investment bank found out, this would be highly impractical for the majority of players in the financial system looking to profit from these contracts. It is reported that one trader forgot to sell on one of their forward contracts with cows as the underlying asset and subsequently was forced to take delivery and leave the office to sort out the cattle storage. To avoid this situation happening each time, there are a few ways that profit can be made without taking delivery:

- Buying a contract with the opposite characteristics (e.g. if a trader is long they can close the trade by opening a short position for an equal amount of the underlying asset) to secure an exit strategy.
- Only entering cash settled contracts.

Most of the time, a bank will close a forward contract position by either entering an equivalent short contract if they are long or an equivalent long contract if they are short. This moves the responsibility outlined by the initial contract onto another party so that the bank only receives cash incoming and outgoings. For example, if a trader is long on 1000 ounces (oz) of gold forward contract, they will close the trade using a short position on a 1000 oz. gold forward contract. When the initial long position expires, they get delivery of the gold (after paying the amount outlined in the first contract) which they then need to deliver to another party whom they made the second closing position with (who will then pay the trader for the gold delivered at the price specified in the second short forward contract).

For most traders this can be inconvenient so exchange traded futures contracts are normally **cash settled**. A cash settlement is a settlement method used in certain futures and options contracts where the seller of the financial instrument does not deliver the actual underlying asset but instead transfers the associated cash position. This cash settlement enables investors to trade more often as the contracts require less time and cost

to deliver upon expiration, therefore resulting in more liquidity being bought into derivatives markets by this mechanism.

Now that you have a firm understanding of how forwards and futures contracts work, we can start to look at the payoff characteristics of these derivatives.

Payoffs

The term 'payoff' is one that you'll frequently come across when reading derivatives. We're going to explain the concept of payoffs separately for each derivatives contract that you see in this lecture as we feel that each derivative behaves differently, so trying to do a 'one size fits all strategy' would make things confusing.

A payoff in the context of forwards and futures is simply the amount of money an investor has lost or gained by preparing a long or short position on a futures or forwards contract – it's a bit like gross profit. Forwards and futures contracts have the same linear payoff characteristics. When we mean linear, the payoff is literally just a straight line. To start, we will look at how a *forward* contract behaves assuming that the contract is at expiry and the contract is cash settled (these payoffs also apply to delivered contracts but the profit is not realised in cash - i.e. you instead own the asset for cheaper/more expensive than without the contract) . When we say the payoff has been 'realised', we simply mean that the profit (or loss, noting it as negative profit) has been confirmed and has been accounted for in the investor's cash flow statements.

In general, the payoff from a **long** position in a forward contract on one unit of an underlying asset is

$$P = S - K$$

where P is the payoff realised, S is the spot price of the underlying asset at maturity of the contract and K is the delivery price (the price to be paid for the asset upon delivery outlined in the forward contract). Similarly, the payoff from a **short** position in a forward contract on one unit of an asset is

$$P = K - S$$

noting that the investor would be paid the delivery cost K and have to buy the asset back at S.

Because it costs nothing to enter into a forward contract, the payoff from the contract is also the trader's total gain or loss from the contract.

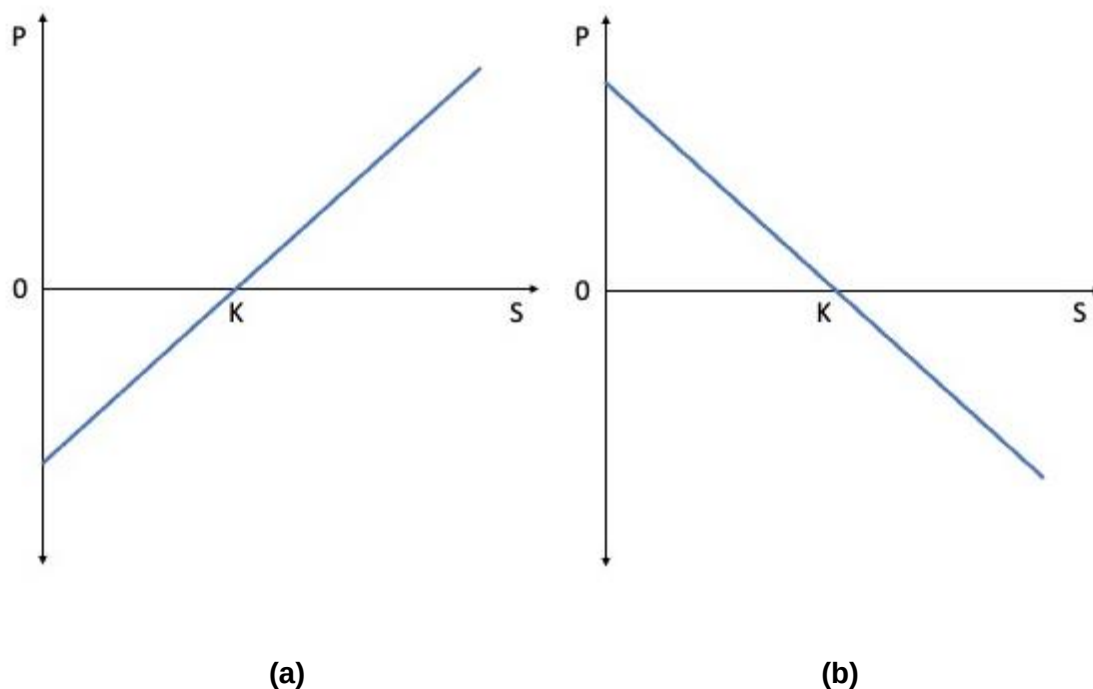


Figure 2: Payoffs from forward contracts: long position (a), short position (b).
 S = spot price at delivery (contract maturity), K = delivery price

Pricing

So, now you have a firm grasp of how futures and forwards work as well as how they differ, there is one key part missing... How are they priced? Both futures and forwards have prices which are heavily reliant on the current price of the underlying asset – think back to what we considered as our model of a derivative. Standardised, exchange traded contracts such as futures are *normally quoted in specific units* where the final cost of the contract is a multiple of this base unit. For example, cattle futures are quoted normally in terms of cents per pound of cow, or oil futures are quoted in terms of price per barrel. The actual contract however specifies how many of this base unit must be purchased upon expiry, e.g. 40,000 pounds of cattle or 1000 barrels of oil.

There are a number of factors that will determine the price of a futures contract, most importantly the spot price of the underlying asset when the contract is entered. Other factors listed in the table below also have a role in determining futures prices as well for different underlying assets.

Factor	Comments
→ Spot price of the underlying asset.	The MOST important factor.
→ Interest rates → Dividend income → Storage costs → Risk-Free rate → Convenience yield	They don't apply to all types of underlying assets as the types of ULs can vary, but they cover (broadly) most of the ones you'll see in the world of finance.

There are a few key ideas to understand here, but don't be worried if these concepts aren't immediately clear as it often takes a while for some mechanisms to sit comfortably before considering more complex trade ideas! Just quickly we will cover how the interest mechanics work and then look at some pricing examples.

Interest Conventions

As most of you may know, we can find simple interest (i.e. no compounding) as

$$P = X \times (1 + t \times r)$$

Where r is the interest rate over one period, X is the value initially invested and P is your final payout after t periods. You may also be familiar with compounded interest, found as

$$P = X \times (1 + r)^t$$

This is where the gain over each investment period is then reinvested and so earnings also start bearing interest. In the financial markets there are odd conventions whereby spans shorter than a year are calculated using simple interest but spans over a year are found using compound interest. This has something to do with how lending parties either apply interest annually, or if the term is shorter than a year, then they apply interest at the end of the term. With this in mind, we can look at how interest rates impact futures prices.

We're going to be talking about the following 4 examples listed below, so familiarise yourselves with them thoroughly as on the next page, we'll be diving deeper to discuss the intricate mechanics behind each of these examples in finer detail:

1. **EXAMPLE 1, Page 21**

The futures price of a non-dividend-paying and non-storable asset (e.g a stock that pays no dividends) is the function of the risk-free rate (or interest rate if we're talking about currencies), spot price and time to maturity.

2. **EXAMPLE 2, Page 23**

Assets that are expected to pay an income through dividends will decrease the price of the futures.

3. **EXAMPLE 2, Page 24**

Storage costs always increase the futures price as the seller of the futures incorporates the cost into the contract.

4. **EXAMPLE 3, Page 24**

Convenience yields, which indicate the benefit of owning another asset rather than the futures, decrease the futures price.

Before we look at how to calculate futures delivery prices, it is important to understand why the above mentioned factors actually impact the prices paid for a futures contract. As with the rest of the market, futures prices are driven by supply and demand. The current price presented at an exchange represents how much the highest bidder will pay. The reason that something such as an interest rate will impact this amount is because markets are generally very efficient in ensuring there are no *arbitrage* (*'riskless return'*) opportunities.

Arbitrage is the process of finding a guaranteed profit. That's the financial definition in terms of what benefits arbitrage can reward you with, but more specifically arbitrage occurs when the same asset is sold in two different places at two different prices. Think of something as simple as a stock. Carnival Corporation & plc is traded on the London Stock Exchange (CCL:LSE) and on the New York Stock Exchange (CCL:NYSE). In theory (noting all prices as USD), CCL:LSE could trade at \$20.00 (that's £16.66 in GBP) whilst CCL:NYSE could trade at \$20.50 at any one moment in time. Noting the \$0.50 difference, a trader could buy 100 shares of CCL at the LSE and sell them at the NYSE and pocket a cool \$50 (0.50×100 , noting we trade 100 shares) profit in difference before costs are paid for – this is achieved simply by buying in one place at a cheaper price and selling it elsewhere at a more expensive price. It's rare for this to occur in stock markets as algorithmic trading is faster than any human trader could mash a stock ticker into Bloomberg.

Let's illustrate arbitrage in the finance world. For example (ignoring interest for now), if we imagine hypothetically that an Apple stock (NASDAQ:AAPL) futures contract expired tomorrow and the delivery price was \$110 but the spot price *for the stock* right now was \$100. We could buy Apple stock now:

Losses	Gains
– \$100	+ 1 Apple Share

Then, we short (remember, sell) the futures contract. Now, when we wait until tomorrow and the contract expires, the following occurs:

Losses	Gains
– 1 Apple Share	+ \$110 (This is how much the buyer of the forward contract pays us, the seller, for 1 Apple share as agreed)

So, now we are left with no apple shares and \$10 out of thin air. See how easy it is to make free money! Well, lots of traders in the market did and now the competition to find these trades is incredibly fierce, it would be almost impossible for a retail trader to find an arbitrage opportunity making even \$0.01 on the above example. This is because each time someone executes the arbitrage trade, they sell the futures contract so its highest available bid decreases (price decreases) and the price of the shares increases and closes the price gap between the two. This happens continuously until there is no arbitrage opportunity left to take and the price stabilises. In this case, there are no factors impacting the price other than the underlying asset (Apple stock) price, therefore the futures contract would settle at a price of \$100.

Now that you understand why futures prices move to consider certain factors, let's look at some arbitrage examples which explain how each type of underlying asset may allow the futures contract to settle at a certain price.

Example 1: Pricing non-dividend-paying non-storable asset futures

Let's revisit the above example with the Apple stocks but now consider the impact of interest rates. Consider the same Apple stock that pays no dividends (in reality, the dividend is 0.63% but ignore this for now) and is worth \$100. We also know that you can borrow or lend money for 1 year at 5%. What should the 6 month futures price of the stock be?

The answer is the initial \$100 with simple interest applied monthly for 6 months at an annual interest rate of 5%:

$$P = \$100 \times \left(1 + 0.05 \times \frac{180}{360}\right) = \$102.53$$

Here we use the annual interest rate multiplied by the number of days where interest applied over the base. The base is the quoted number of days for the given interest rate of 5%.

Be careful when assuming a base of 365 days, this may seem like common sense, but if working with most interest rates the base is actually 360. This is a very odd convention with the UK being part of only a handful of countries to use 365 days.

More generally, the formula we use is given as:

$$P = S \times \left(1 + r \times \frac{D}{B}\right)$$

where P is the futures price, S is the spot price of the underlying asset, r is the interest rate achieved (quoted as an annual amount without compounding), D is the days until trade expiry and B is the base used (usually 365). Note that the **above equation only applies for periods under a year!** If we instead wanted to find the future value for a period over a year, we would use the following formula which assumes compounded interest

$$P = S \times (1 + r)^t$$

Think about what a trader would do if the futures price was *lower* than the spot price for a stock that they owned – this is known as futures contract mispricing. Understanding why the futures delivery price increases with positive interest requires us to consider e.g. if the quoted futures price was instead \$95. In this scenario, an investor who already owns the stock could sell it for \$100 and enter into a long forward contract to buy it back for \$95 in 6 months. The proceeds of selling the stock could be invested at 5% to earn \$2.47 (as seen above – the same calculation) and after a year, the investor would pay \$95 to get the Apple stock back. This process would guarantee a profit of $\$100 - \$95 + \$2.47 = \7.47 despite any moves in the underlying stock.

This is summarised in the table below, showing the relevant cash flows and overall total profit.

Event	Net Cash Flow (\$)
1. Short the underlying AAPL stock.	+ 100.00 + (you owe 1 Apple share)
2. Invest the proceeds i.e. loaning it at 5%	$100.00 \times (1 + 0.05 \times 180/365)$

interest rate for one year, monthly compounded.	= 102.47 i.e. + 2.47
3. Long a share of AAPL 1 year later at \$95 by exercising the futures option.	– 95.00 + (you receive an apple share and immediately give it to the person you owe)
TOTAL PROFIT	+ 100.00 + 2.47 – 95.00 = 7.47

As we learned earlier, this is an arbitrage opportunity and in reality, the price would move to ensure this would not occur and the future contract is correctly priced. At a futures price of \$102.47, the invested income from selling the stock in steps 1,2 would be cancelled out by the higher delivery price. i.e. + \$100 from selling the stock + \$2.47 interest - \$102.47 delivery price = \$0. Note here that the reason why I get \$0 overall is that I sell the underlying stock today at \$100 **and** exercise a futures contract at the same time from another seller. I invest the \$100 I earned to earn another \$2.47 thanks to the magic of interest, but then I will end up buying back an AAPL stock through the futures contract at a price of \$102.47, leaving me with... nothing. This is a simple example although arbitrage works more commonly when sources of credit e.g. loans are involved.

If the price of the futures contract was instead higher than \$102.47, e.g. \$110, the reverse operation could be used to make risk-free profit from arbitrage.

1. We could sell the forward contract for e.g. \$110 (such that we owe one share in 6 months time). Here no money is moved immediately.
2. Borrow money from a bank and buy the underlying asset (AAPL share) at its spot value for \$100.
3. 6 months later, we then transfer this AAPL stock to the buyer and pocket the \$110 for the price of the futures contract. We return the \$100 we borrowed, along with an additional \$2.47 interest rate to give us an overall profit of \$7.53.

Event	Net Cash Flow (\$)
1. Sell a futures contract for 6 months time at a price of \$110	0 - no money moves yet
2. Simultaneously borrow \$100 from a bank and buy an AAPL share at the spot price of \$100	- \$100 + 1 AAPL share
3. 6 months later, transfer the AAPL stock to the buyer of the futures contract you sold and receive \$110. Here you would also pay the interest due on the initial loan.	- 1 AAPL share + \$110 - \$100 (loan) - \$2.47 (interest)
TOTAL PROFIT	- 100.00 + 110.00 – 2.47 = 7.53

Tricks to understand futures pricing

Another mindset which can help you to understand futures pricing is to think:

If I have \$100, *what would I prefer as a trader, to receive a set asset now for \$100, or to receive the asset in the future for \$100?* **An accurate price will make this decision impossible.**

- *If I receive the asset now for \$100, in one year's time I'm left with no money and the asset (let's say a bar of gold for example)*

OR

- *If I have to only pay \$100 in the future I can simply invest my \$100 now, earn interest, and then buy the asset in the future for \$100. This leaves me with the asset, plus some extra income in one year's time.*

The second choice here is obviously more appealing, therefore we expect that we would have to also lose the potential interest if purchasing the asset in the future (through an inflated futures price) for both outcomes to be equal.

A futures price is always the future value of the underlying asset if it was bought today

Example 2: Pricing income yielding and storable asset futures

Let's take our model from example 1 a bit further. Again, we can build on this model by including any dividends that Apple may pay to its shareholders. It may not seem intuitive but stock dividends do the opposite to what interest rates do in terms of the value of our futures contract. First, we're going to present to you the equation used to price income-yielding futures contracts and then justify why it is written the way it is. So, we can incorporate dividends into the above equation as

$$P = S \times (1 + (r - d) \times 180/365)$$

Please be aware that this equation only applies to periods below a year - a derivation of the pricing equation for periods over a year is given in the slides

For time horizons over a year, the pricing equation is given as

$$P = S \times ((1 + r)^t - (1 + d)^t - 1)$$

The derivation of this equation can be found in the slides

where d is the percentage (%) dividend paid on the stock per year. This formula assumes that the dividend is paid out semi-annually, such that in this scenario the dividend compounded income can be neglected. The

futures price decreases when there is a known interest income because the long side buying the futures does not own the asset and, thus, loses the interest benefit. Otherwise, the buyer would simply buy the underlying asset for a year instead of the forward/future. *Why would you buy a futures contract in AAPL and wait a year when you could just buy the AAPL stock itself and benefit yourself to the 0.63% dividend rate that it offers.* In the case of stock, the long side loses the opportunity to get dividends by entering a forward so a compensation must be made in the forward/futures price to deter traders from instead buying the underlying stock.

Now that we've 'priced in' the dividend rate, one more important aspect in pricing considers assets with storage costs. The long side does not incur any storage costs until it actually owns the asset. Therefore, the short side charges the long side for the compensation of storage costs in the futures price. This makes sense – someone has to pay the warehouse for area to store the underlying asset. If the futures price was not increased to account for storage, it would be far more beneficial to enter the futures contract rather than buy the asset now and store it for a year. Storage costs are usually given as a single figure per given amount of storable asset rather than a percentage, therefore our pricing mode for periods below a year can be adapted as:

$$P = S \times \left(1 + (r - d) \times \frac{180}{360}\right) + k \times t$$

where k is the cost to keep (store) the underlying asset per unit time and t is the number of units of time until contract expiry. All commodities require storage and therefore often have higher futures prices than their spot prices.

Example 3: Pricing FX futures

Some of the most commonly traded types of futures contract are based on currency pairs (e.g. EUR/USD). As currencies don't require storage (because they're electronic, unless you want to take delivery of bank notes or coins) and provide no income such as dividends, the futures price is impacted only by the interest rates which apply to each currency.

As we found in prior lectures, buying a currency pair requires borrowing the second currency and using it to buy the first currency, e.g. for a long position in EUR/USD, you would borrow USD and exchange it for EUR. Therefore, we earn interest on our EUR and pay interest on our USD.

Now imagine that we want to buy a 2 year forward for GBP/USD, how would we price it? We know that the current spot price of GBP/USD is 1.18, i.e. 1 pound exchanges for 1.18 dollars, and the annual US interest rate is 2.50% and the UK rate is 1.75%.

Remembering that a futures price is always the future value of the underlying asset if it was bought today, we can calculate the future outcome of buying this currency pair today given that the exchange rate doesn't move.

We will be buying pounds and so, at the current interest rate, £1 now will be worth $(1 + 1.75\%)^2 \times 1 = £1.0353$ in 2 years time. Conversely, we will be borrowing dollars and so, at the current interest rate, \$1.18 borrowed now will be $(1 + 2.5\%)^2 \times 1.18 = \1.2397 owed in 2 years. Therefore, our debt in 2 years will be worth \$1.2397 and our assets will be worth £1.0353. To avoid any arbitrage opportunity, this is what we must therefore pay for the 1 year maturity forward contract.

This leaves our quoted exchange rate as $\frac{\$1.2397}{£1.0353} = 1.1974$.

It is common to find this kind of thought process very confusing so one way of finding the price of a forward or future can be to find the future value of the underlying asset as if you had instead bought the asset at spot value, i.e. today on the market and waited until the maturity of the forward/futures contract.

The general formula for finding the forwards price of a given currency pair is given as

$$F = S \times (1 + r_2)^t / (1 + r_1)^t$$

where F is the quoted currency pair ratio, S is the current spot exchange rate, r1 is the interest rate which acts on the first currency in the pair (the one being bought) and r2 is the second currency in the pair (i.e. the one being borrowed). t represents the number of time periods that the interest acts over. This equation can be used for positions held over a year, for shorter positions a similar simple interest version can be used,

$$F = S \times (1 + r_2 \times D/B) / (1 + r_1 \times D/B)$$

Where D is the number of days until expiry and B is the base (e.g. 360 days in US and 365 days in the UK - odd convention I know!)

Typically, futures contracts have higher settlement prices than their spot prices, therefore the further away a contract expiry, the more expensive it is. This is known as **contango**. Under some conditions, however, the opposite situation might occur, and the futures price could be lower than the spot price. This could be due to a temporary scarcity of the commodity on the spot market, for example, due to a restricted current supply, making the price of the commodity unusually high and expected to decrease in the future - why would I agree to buy a barrel of oil in a year's time at today's price if I know the price is going to decrease? This situation is known as **backwardation**. If the market is in backwardation, the fair value of a futures contract is more difficult to calculate.

The examples that we have looked at so far find the fair value of futures and forwards contracts at any time before the contracts expires. Using the above equations we can also see that as $t \rightarrow 0$ the contract prices become equal to the spot price of the underlying asset. We can see the relation between the contract and spot price as the contract approaches maturity in Figure 3 below.

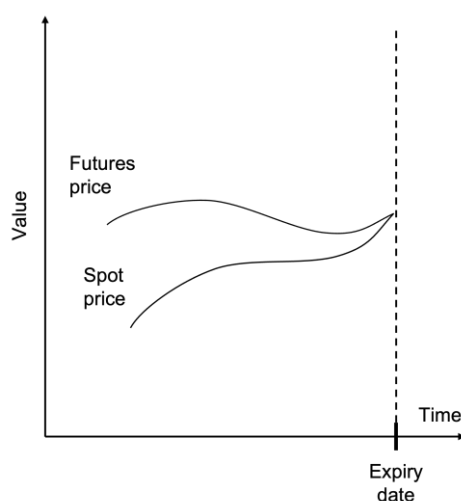


Figure 3: Spot price vs futures price as the contract approaches maturity